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Session Date: May 26, 2026 (1 hour)

Objective: Run timing experiments for full vs incremental mode and answer discussion questions.

Attempts made:

Ran both modes with TC=2 TQ=2. Both diffed clean against expected_queries.txt so correctness is confirmed for both.

Solutions/Workarounds:

No issues this session.

Commits: add phase 4 journal sessions 1-4

Experiments/Analysis:

Mode	Wall time
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Full	30898.7 ms
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Incremental	13707.1 ms
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Incremental is about 2.25x faster. Full mode resets `dist[]` to INF at every query and runs BSP from vertex 0 across the whole graph. Incremental reuses the previous `dist[]` and only re-relaxes from the dirty frontier — source vertices of ADDs since the last query — so most of the graph doesn't get touched at all.

Removing `WaitUntil`: without it a query can start before all ADDs with version $< q$ are applied. The output would be wrong and non-deterministic since it depends on thread scheduling. The edge version filter doesn't help here — it hides future edges but not ones that are still in flight.

TC/TQ combinations on incremental: with only 5 queries the SSSP is the bottleneck. Increasing TQ from 2 to 4 helped more than increasing TC since updates are cheap compared to SSSP. Going above TQ=4 flattened out due to thread overhead per BSP round.

Learning outcome:

Incremental SSSP only works here because the graph is add-only and distances can only decrease. The dirty frontier is always a valid starting point since any vertex not in it already has the correct distance for this query's version cutoff.